Practice Sheet 3: Functions and Calculus

Name: _____ Class 12 Date: _____

Section A: Graphs

1. How does the graph of $y = a \cdot x^2$ change as a changes?

2. Find the slope of the curve $y = x^2$ at the point $x = 1$.

3. Find the area under $y = x^2$ between $x = 0$ and $x = 2$.

4. Why is the derivative of $\sin x$ equal to $\cos x$?

Section B: Algebra and Integration

5. Solve $x^2 - 5x + 6 = 0$.

6. Find the integral of $x \cdot e^x dx$.

7. Simplify $(x^2 - 9) / (x - 3)$.

Section C: Check my working

Each answer below has a mistake. Circle a line and ask if it's right.

8. Solve $2(x + 3) = 14$

$2x + 3 = 14$

$2x = 11$

$x = 5.5$

9. Differentiate $y = (3x + 1)^2$

$dy/dx = 2(3x + 1)$

10. Find the integral of $2x\,dx$

$= x^2$

11. Expand $(x + 4)^2$

$= x^2 + 16$